01-MICROCONTROLLERS-AND-DEV-BOARDS

Item #	Component / Board Name	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**ESP32 DevKit V1 (30-Pin / 38-Pin)**	32-bit Dual-Core Xtensa LX6 @ 240 MHz; onboard 2.4 GHz Wi-Fi + Bluetooth BLE 4.2/5.0; 520 KB SRAM; 4 MB Flash; 3.3V logic; capacitive touch, ADC, DAC, PWM on all GPIOs	**16 Units**	**Consumable**
02	**Seeed Studio XIAO ESP32-S3 Development Board**	WiFi & Bluetooth 5.0 Microcontroller Board with integrated battery charging; ESP32-S3 dual-core Xtensa 32-bit LX7 @ 240 MHz; ultra-compact thumb-sized form factor; ideal for IoT, robotics, wearable & smart device projects	**5 Units**	**Consumable**
03	**Espressif ESP32-S3-DevKitC-1-N8R8 Development Board**	Flagship ESP32-S3 development board featuring ESP32-S3-WROOM-1 module; 8MB Octal SPI Flash + 8MB Octal PSRAM; 2.4 GHz Wi-Fi (802.11 b/g/n) + Bluetooth 5 (LE); dual USB-C ports (OTG + UART); hardware vector instructions for TinyML/Edge AI	**5 Units**	**Consumable**
04	**Arduino Uno R3 (or equivalent)**	ATmega328P 8-bit AVR; 5V; 14 Digital I/O (6 PWM); 6 Analog inputs; 16 MHz; includes protective casing and USB Type-A to Type-B programming cable	**30 Units**	**Consumable**
05	**Arduino Nano (or equivalent)**	ATmega328P 8-bit AVR; 5V; breadboard DIP form factor; 14 Digital I/O; 8 Analog inputs; Mini-USB cable included	**5 Units**	**Consumable**
06	**Arduino Mega 2560 (or equivalent)**	ATmega2560 8-bit AVR; 5V; 54 Digital I/O (15 PWM); 16 Analog inputs; 4 UART serial ports; USB cable included	**5 Units**	**Consumable**
07	**Raspberry Pi 3 Model B+**	Quad Core 64-bit Broadcom ARM @ 1.4 GHz; 1 GB RAM; Wi-Fi & BLE; 40-pin GPIO; 4x USB; HDMI; 5V 2.5A power supply & protective enclosure	**4 Units**	**Consumable**
08	**Raspberry Pi 5 (2GB RAM)**	64-bit Quad-Core ARM Cortex-A76 @ 2.4 GHz; 2 GB LPDDR4X SDRAM; VideoCore VII GPU; Dual 4Kp60 micro-HDMI; PCIe 2.0 interface; Gigabit Ethernet, Dual-band 802.11ac Wi-Fi, Bluetooth 5.0 / BLE; 2x USB 3.0, 2x USB 2.0; USB-C 5V 5A PD input	**2 Units**	**Consumable**
09	**Raspberry Pi 4 Model B (2GB RAM)**	64-bit Quad-Core ARM Cortex-A72 @ 1.5 GHz; 2 GB LPDDR4 SDRAM; Dual 4Kp60 micro-HDMI displays; Gigabit Ethernet; 2.4/5.0 GHz Wi-Fi, Bluetooth 5.0; 2x USB 3.0, 2x USB 2.0; USB-C 5V 3A input	**2 Units**	**Consumable**
10	**Raspberry Pi Zero (v1.3)**	Single-Core ARM11 @ 1 GHz (BCM2835); 512 MB LPDDR2 RAM; mini-HDMI video; micro-USB OTG data port; micro-USB power; CSI camera connector; ultra-compact 65×30mm board size	**2 Units**	**Consumable**
11	**Raspberry Pi Zero W (with WiFi)**	Single-Core ARM11 @ 1 GHz; 512 MB RAM; onboard 802.11 b/g/n Wi-Fi + Bluetooth 4.1 / BLE; mini-HDMI; micro-USB OTG port; CSI camera connector; ultra-compact form factor	**2 Units**	**Consumable**
12	**Raspberry Pi Zero 2 W**	64-bit Quad-Core ARM Cortex-A53 @ 1 GHz (RP3A0 SiP); 512 MB LPDDR2 RAM; onboard 2.4 GHz Wi-Fi + Bluetooth 4.2 / BLE; mini-HDMI; micro-USB OTG; CSI-2 camera connector; up to 5x faster performance	**2 Units**	**Consumable**
13	**HIKVISION Extreme 16GB microSDXC Memory Card (Class 10, V30)**	High-speed 16GB Class 10, V30 MicroSD card; 92 MB/s read speed, 15 MB/s write speed; compatible with smartphones, cameras, CCTV; drop protection; 3-year warranty; **Dedicated for Raspberry Pi 3B+ Single-Board Computer (Operating System Boot & Local Storage)**	**5 Units**	**Consumable**
14	**NodeMCU v3 (ESP8266)**	ESP8266 Wi-Fi SoC with onboard CP2102/CH340 USB-UART; 3.3V logic; integrated TCP/IP stack; PCB trace antenna	**10 Units**	**Consumable**
15	**WeMos D1 R2 (ESP8266)**	Arduino Uno form-factor Wi-Fi development board powered by ESP8266; 5V–12V barrel jack input; 11 Digital I/O; 1 Analog input	**5 Units**	**Consumable**